

Passive Monitoring of Fall-Events Using Floor Vibrations

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1 Summary

This system acts as a prototype of a floor-mounted sensing system that alerts in the event that a person falls by monitoring the vibrations resulting from the large impact. An accelerometer detects any patterns within the wave's amplitude that coincides with the observations from a 'fall-event'. A loudness sound sensor is used alongside this to increase confidence in these alerts, though it was found to have a too low accuracy to detect falls alone. This is able to distinguish falls at close range to the sensors; however, this prototype is not sufficient to monitor events at distances. Machine learning is needed to classify a fall from other daily activities and increase this range to estimate the infrastructure needed to successfully monitor a 'smart room' within a household.

2 Background and Motivation

For those who live alone, there is often great concern about the possibility of a life-threatening accident, and a contingency plan is needed to raise an alarm for help as otherwise they are often left undetected for some time [1]. This is especially true for the elderly or those who are fall-risk, as there are many hazards within a household that increases the likelihood of a fall [2]. Even relatively low-impact falls, which do not lead to loss of consciousness or heavy head-trauma, may be a serious emergency for those who are unable to right themselves. This is one of the greatest consequences of a fall, as those left immobilised on the ground are at high risk of long-lasting health conditions that could lead to loss of life. There is a great importance for early detection devices that automate timely alerts to those who can attend and offer assistance [3].

3 Related Work

Within a research review, Yu classifies the different approaches of fall-detection systems into three categories: wearable devices, camera-based methods, and ambient device systems, as shown in Figure 1.

3.0.1 Wearable systems

Early systems equip the user with a personal alarm system worn around the wrist or neck, to manually send alerts after a fall. This requires the device to be within reach and is unhelpful if the person becomes unconscious from the head trauma [1]. Automated detection is available through in-built sensors such as accelerometers and gyroscopes. However, this relies on the assumption that the device will be worn reliably everyday, this is not always the case, especially if the user is feeling well and does not anticipate a fall [4].

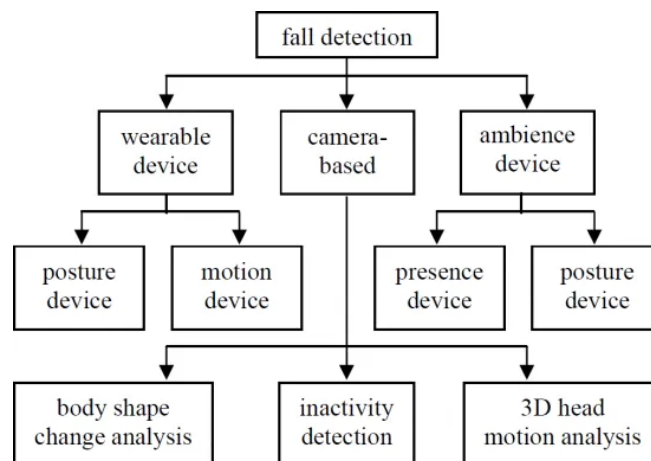


Figure 1: Approaches to sensor-based fall detection from 'Approaches and principles of fall detection for elderly and patient' [4].

3.0.2 Ambient devices

An alternative vision-based approach within a “smart room” allows for the recognition of fall characteristics [5]. For example, pyroelectric IR sensor arrays can be used to estimate positioning of a person based on their thermal patterns [6]. Ultrasonic rays have also been used to express a fall by the changes of distance between the person and the sensor [5]. However, these require classification between fall events and different daily activities, such as sitting or bending down, that will have very similar data patterns [6].

Zigel et al. [7] present a machine-learning based system that uses floor vibrations to distinguish the impacts of a “human mimicking doll” from daily activities. Similarly, Alwan et al. [8] used this and a piezo sensor to compare the vibrations of a simulated fall to the dropping of objects of much smaller weights. Our project builds upon these systems to determine how the material of the surface impacts the prediction accuracy.

4 System Design

This project focuses on a stationary prototype that monitors the environment for any large surface vibrations that occur in the event of a fall. Several small sensors will be placed alongside the perimeter of a room, to continuously listen for a sharp peak that suggests a large impact has taken place. Like many ‘Smart Home’ devices, this system relies on the user’s implicit interactions and contextual knowledge to infer and categorise the data collected from the external sensing of the moving body [9]. As the user does not interact with the system through a traditional interface, it is necessary to consider the challenges raised by Bellotti’s design framework [10].

There is a need to communicate to the user that their actions are being monitored as the sensors will be ‘always on, always listening’. This raises privacy concerns within the home, though is less invasive than cameras or body monitoring [5]. The user must be able to avoid interaction with the system through an external module that does not require the user to walk up to switch it off- and be captured in the process. It is likely that a larger ‘control’ module will be needed, which will monitor the sensors themselves and allow the user to stop the data capture. Due to the severity of missing a true event within a healthcare system, a validation module allows for user-labelling of data to manually trigger missed alerts and cancel false events. This section of the expanded system, shown in Figure 2, would have different evaluation priorities to minimise the rate of false-negatives to avoid erroneously cancelling a true event.

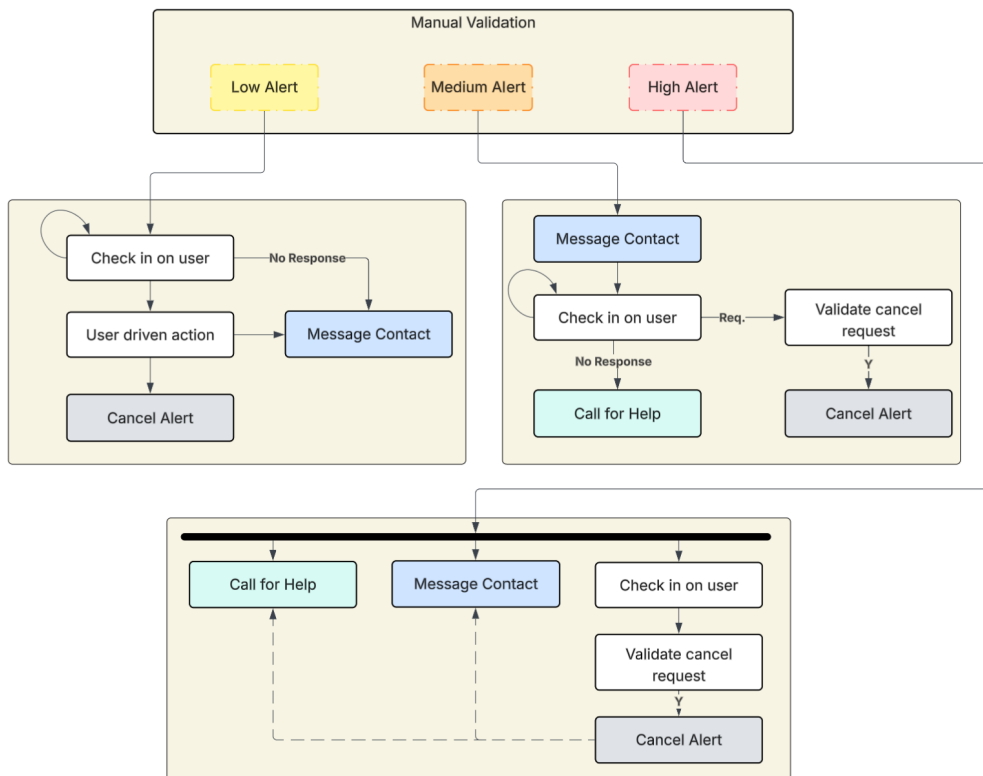
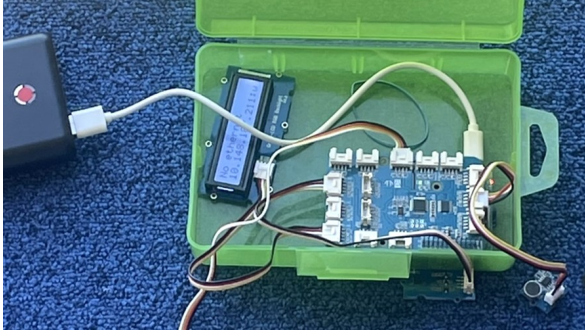


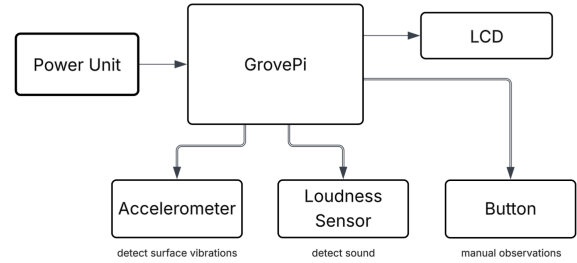
Figure 2: The validation module within the prototypes extended system, to alert a user’s care-givers or emergency services if there is a strong likelihood a fall has taken place.

5 Implementation

A fall-event is simulated through regular footsteps around the sensed area and a single large impact, to mimic noise from daily-activities that would happen around the sensors, and to ensure that these can be distinguished from a fall.



(a)



(b)

Figure 3: GrovePi sensors used to collect raw data.

5.1 Capturing Data

The sensing system used to collect the training and testing data is shown in Figure 3. A button manually logs the beginning and end of each event and the accelerometer data within these time periods is observed to infer the ground-truth labelling and identify the relevant points within the more obscured sound data. As seen in Figures 4, 5, three separate positive events have been simulated to build the prototype.

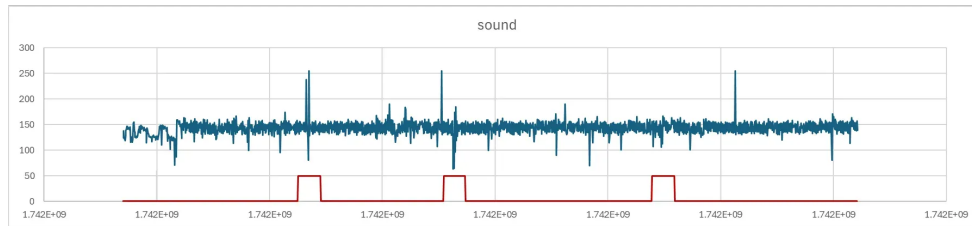
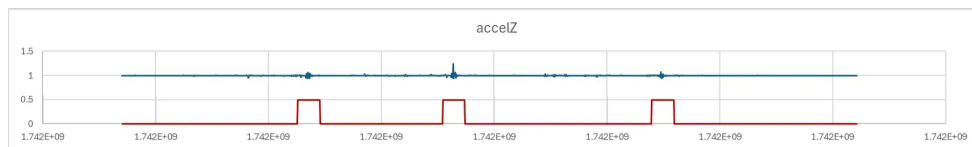
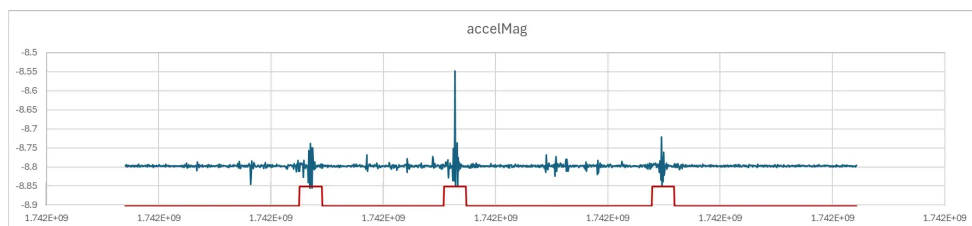


Figure 4: Recorded data from the loudness sensor. The final event has been detected with a considerable delay and will be counted as a false positive event for testing purposes.



(a)



(b)

Figure 5: Recorded data from the accelerometer sensor: (a) returned z-value, (b) calculated magnitude - this is independent from rotation and will form large peaks even if not placed flat against the surface.

5.2 Filtering

The filter pipeline in Figure 6 shows how both data streams have been cleaned using linear filters: a low-pass filter is sampled at a slow rate to smooth low-frequency noise, and a high-pass filter is placed on the result to reintroduce the high-frequency values. This process adds latency within events, which can have great repercussions during the healthcare setting where timely detection is a necessity [3].

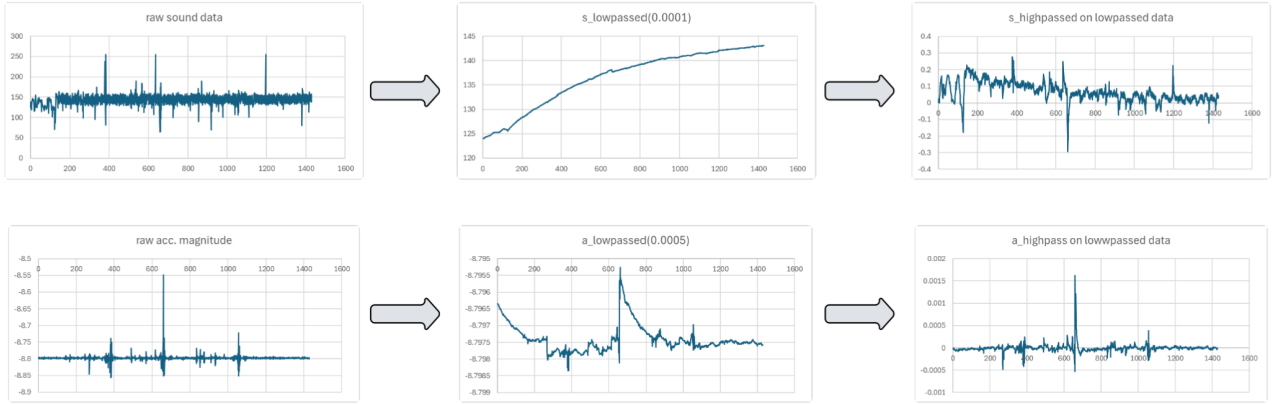


Figure 6: Filter pipeline on the raw sensor data.

5.3 Thresholding

A positive event within the accelerometer is categorised as low-amplitude oscillations in the data signal, followed by several high peaks. An event within the sound data can be seen by a sudden increase in intensity, though this is not always present.

A static threshold would restrict performance, as the intensity of the peaks differs depending on the material of the surface the device is placed upon, and the unpredictability of noise within the environment [7]. In order to be deployed within different rooms of the user's home, adaptive thresholding is used to categorise events based on the relative change in intensity from the previous 50 values within a history buffer. Due to the nature of oscillating waves, thresholded events must be considered in pairs, as the value must rise and fall outside of the average range. For this, two thresholds have been implemented, as shown in Figure 7a. Similarly, a single adaptive threshold is placed on the sound data in Figure 7b.

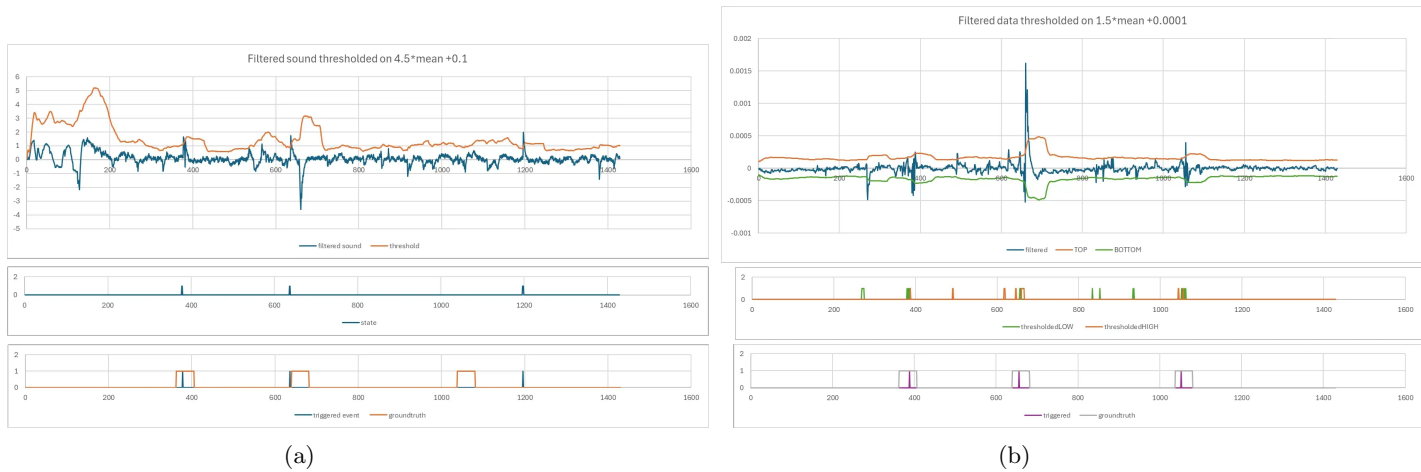


Figure 7: The results from thresholds on sound (a) and accelerometer (b) data, and the inferred events after filtering only thresholded pairs. This has been compared to the ground truth values.

5.4 Combining Sensors

Each sensor is tested independently to measure their respective accuracies and estimate the probability of a true-event corresponding to a positive reading, so that the suitable alert can be raised [11]. Figures 8, 9, visualise the design prioritising the accelerometer results; while the loudness sensor increases the probability of a true-event, it alone is not enough to confidently report a fall.

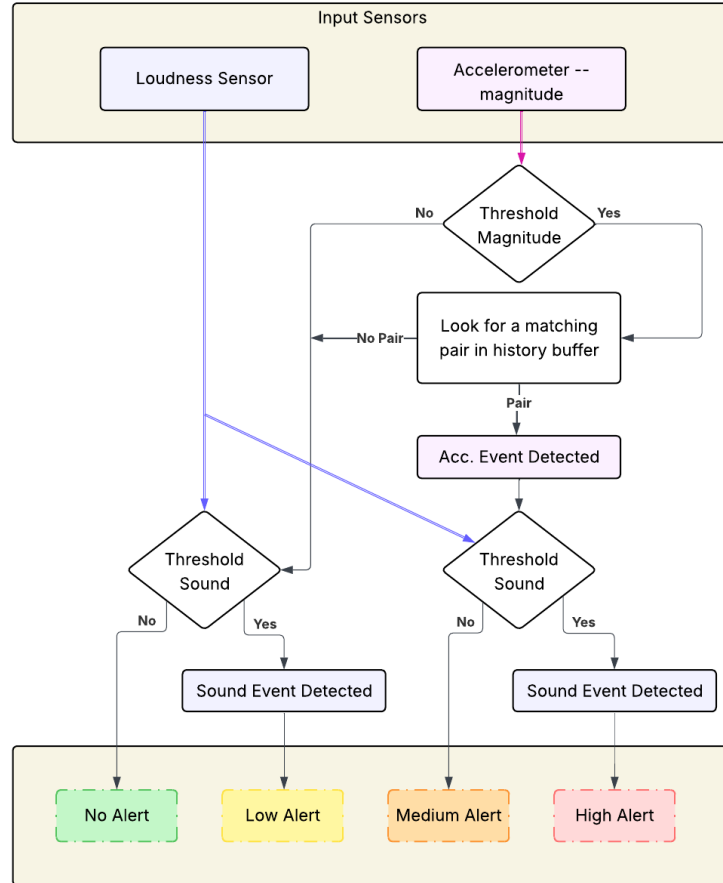


Figure 8: Decision process when combining sensor data to infer fall-events.



Figure 9: The produced alerts when processing the sensor data. The priority of the alert increases as more data supports the probability that the inferred event is true. The final event is given two separate alerts due to the delayed response in the sound data.

6 Testing

The tests are executed remotely through the GrovePi emulator [12]. Calibration and validation tests have taken place within a carpeted room to simulate a living area within the user's home, a sketch of the testing environment is given in Figure 10.

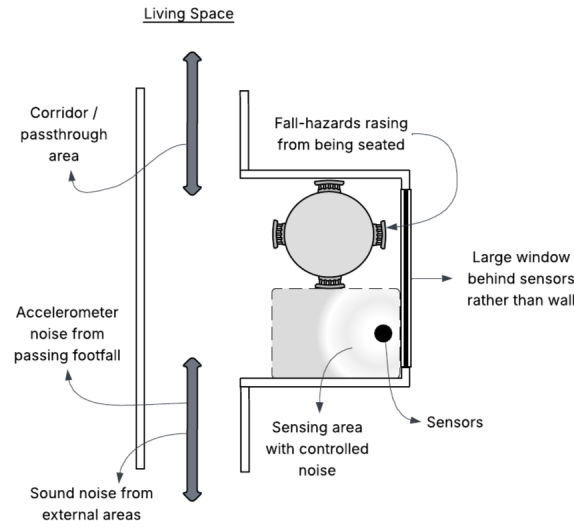


Figure 10: Floor plan of the main testing environment, the sensors are placed in the corner of the room upon a carpeted surface.

6.1 Validation of system

Following the calibration of the filters and threshold values, the system is tested on unseen validation data. In Figure 11 the produced outputs are compared with the observed ground truth and aggregated for each experiment. No significant performance decrease has been found, indicating that the settings have not overfitted onto the calibration data. Graphical visualisation of these results can be found in the Appendix folder.

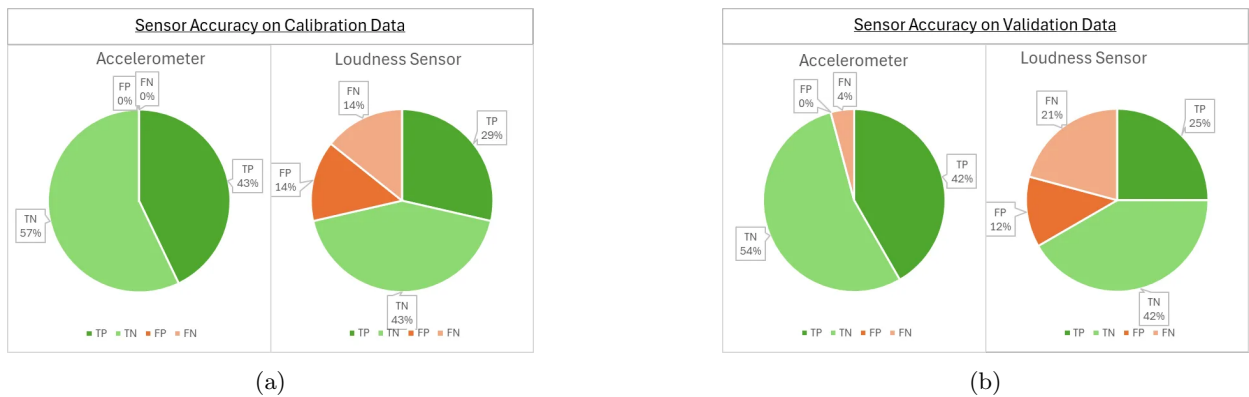


Figure 11: Results from the validation tests within the carpeted room. Three positive events have been performed within the initial data (a), while ten events have been recorded for system validation (b). Green shades indicate that the event state has been predicted correctly, while red shades are erroneous.

6.2 Testing surfaces

Additional tests have taken place on both tiled and laminate flooring to represent other rooms within the house that a user may want to be monitored within; such as a bathroom or kitchen, which have their own fall hazards. Diagrams of each environment setup are given in Figure 12. Visualisation of the results are provided in the Appendix C and D.

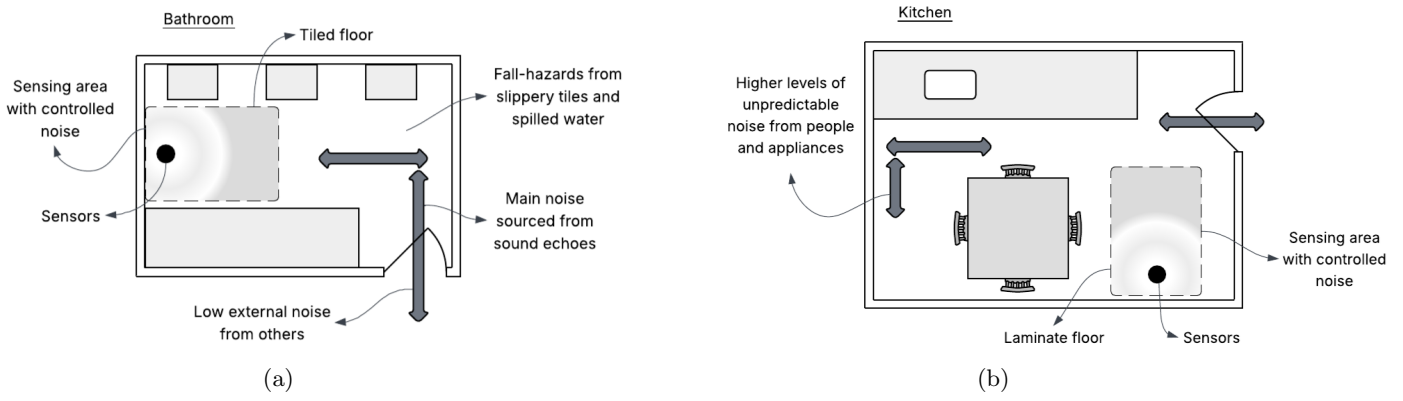


Figure 12: The testing environments used to measure vibrations across tiled surfaces (a) and laminate floors (b).

The percentages of successful recorded events are visualised in Figure 13, and accuracy counts are given in Figure 14. The prototype has been found to adapt well to the tiled surface, while the laminate has caused a significant performance decrease. The data collected from the loudness sensor on the laminate flooring has been overly filtered, causing any peaks from the fall-events to be obscured. This is likely due to the large levels of noise within this environment. The final system would likely require individual calibration based upon the room type it is deployed in, to minimise these dangerous false events.

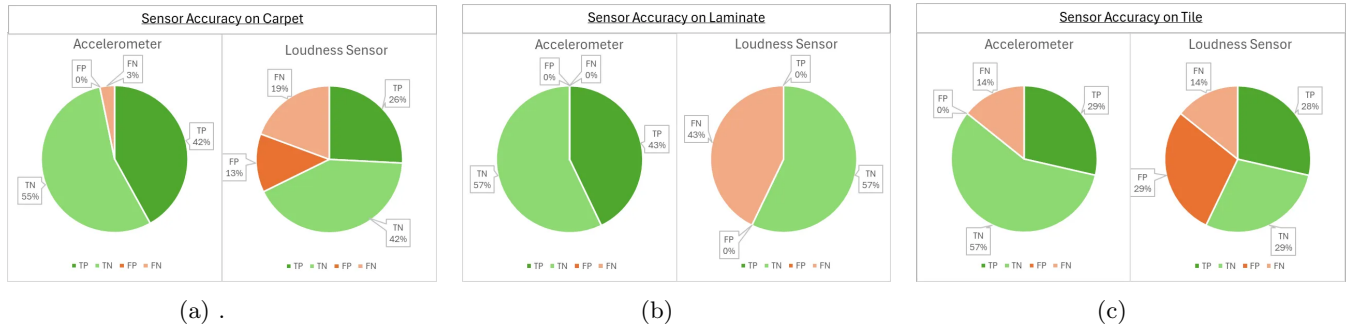


Figure 13: The accuracy of each sensor's inferences made on the tested surfaces: carpet (a), laminate (b), tile (c). The majority of the data has been collected on the carpeted surface.

Accel. Sensor Report	Real Event	
	TRUE	FALSE
TRUE	18*	0
FALSE	2	25

Sound Sensor Report	Real Event	
	TRUE	FALSE
TRUE	10	6**
FALSE	10	19

(a)

Sensor Report (A,S)	Real Event	
	TRUE	FALSE
(0,0)	0	19
(0,1)	2	6
(1,0)	10	0
(1,1)	8	0

(b)

Figure 14: (a) Matrix of results against the observed ground-truth to show the accuracy of each sensor across all tests executed. (b) The combined report for accelerometer A and loudness sensor S related to the truth of the registered event. *' Count includes duplicated events. **' Count includes true delayed events.

6.3 Testing sensor proximity

An additional experiment was conducted to find where vibrations can no longer be distinguished from the noise of footsteps. This allows us to consider the sensitivity of the sensors and the infrastructure needed to successfully detect a fall anywhere within the smart room.

The tests were performed at increasing distances from the sensors, as shown in Figure 15, and the resulting data collected can be found within the Appendix folder. It was observed that events further than one meter have unsuccessful thresholding and a machine learning approach is required to analyse patterns within the frequencies rather than searching for stand-alone peaks.

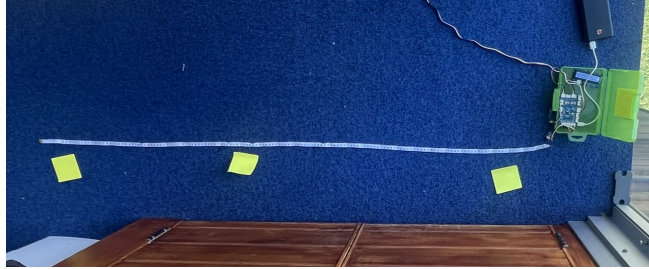


Figure 15: Experiment setup to test sensor performance detecting single impact events at incrementing distances from 20cm up to 1.8m.

7 Reflection

During testing a single short impact has been used to reflect a true event; however, in reality there are many different types of falls that have different impact patterns and force strengths which should be considered. Other work, such as that of Rimminen Figure 16, uses Bayesian filtering to classify these falls from pose estimation [13]. Further tests are needed to explore how the system responds to high-level of noise within the environment, and any interference this brings to the vibration waves.

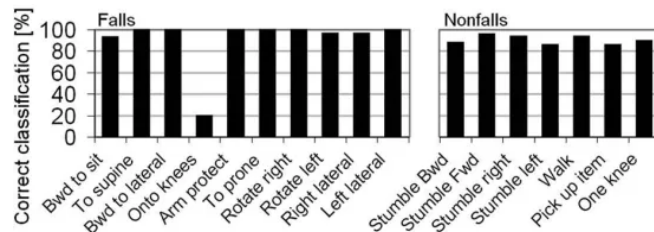


Figure 16: Types of falls to be considered, from ‘Detection of falls among the elderly by a floor sensor using the electric near field’ [13].

7.1 Ethical considerations

While this approach to fall-detection is less intrusive than an image based system, there are concerns of monitoring the user within their home and the discomfort this may bring. As the system waits for a large impact, it will inevitably record vibrations made by the user’s footsteps as they carry out their daily activities. When paired alongside positioning data and contextual information, this could be used to infer the user’s actions and routines within their home. This raises concerns to regulate who will have access to the collected data and potential misuse. Sharing the data with those who are trusted may be beneficial to validate missed events; however, this could be abused through excessive monitoring or using it as a means of control.

There is ongoing research into information that can be extracted from these vibrations, that could be performed without the user’s initial consent. While this would often require highly-sensitive sensors, such as detecting a person’s heartbeat or body motions, identifiable inferences can be made from the higher-level signals [14]. For example, the cadence and walking speed of a person’s gait can be found from their registered footsteps - often used for diagnostics and to personalise healthcare [15]. There is current research into using gait as an “behavioural biometric” to identify a person through computer vision techniques, and the same principles could be applied to features extracted from vibration patterns [16].

References

- [1] Vallabh P. and Malekian R. “Fall detection monitoring systems: a comprehensive review.” In: *J Ambient Intell Human Comput* 9 (2018), pp. 1809–1833. DOI: <https://doi.org/10.1007/s12652-017-0592-3>. URL: <https://link.springer.com/article/10.1007/s12652-017-0592-3#article-info>.
- [2] Nooruddin S., Islam M.M., and Sharna F.A. et al. “Sensor-based fall detection systems: a review.” In: *J Ambient Intell Human Comput* 13 (2022), pp. 2735–2751. DOI: <https://doi.org/10.1007/s12652-021-03248-z>. URL: <https://link.springer.com/article/10.1007/s12652-021-03248-z#article-info>.
- [3] Frédéric Bloch. “Critical Falls: Why Remaining on the Ground After a Fall can be Dangerous, Whatever the Fall”. In: *Journal of the American Geriatrics Society* 60.7 (2012), pp. 1375–1376. DOI: <https://doi.org/10.1111/j.1532-5415.2012.04017.x>. URL: <https://agsjournals.onlinelibrary.wiley.com/doi/abs/10.1111/j.1532-5415.2012.04017.x>.
- [4] Xinguo Yu. “Approaches and principles of fall detection for elderly and patient”. In: *HealthCom 2008 - 10th International Conference on e-health Networking, Applications and Services*. 2008, pp. 42–47. DOI: 10.1109/HEALTH.2008.4600107.
- [5] Chokmongkol Nadee and Kosin Chamnongthai. “An Ultrasonic-Based Sensor System for Elderly Fall Monitoring in a Smart Room”. In: *Journal of Healthcare Engineering* 2022 (Nov. 2022). DOI: 10.1155/2022/2212020.
- [6] Nishat Tasnim Newaz and Eisuke Hanada. “An Approach to Fall Detection Using Statistical Distributions of Thermal Signatures Obtained by a Stand-Alone Low-Resolution IR Array Sensor Device”. In: *Sensors* 25.2 (2025). ISSN: 1424-8220. DOI: 10.3390/s25020504. URL: <https://www.mdpi.com/1424-8220/25/2/504>.
- [7] Yaniv Zigel, Dima Litvak, and Israel Gannot*. “A Method for Automatic Fall Detection of Elderly People Using Floor Vibrations and Sound—Proof of Concept on Human Mimicking Doll Falls”. In: *IEEE Transactions on Biomedical Engineering* 56.12 (2009), pp. 2858–2867. DOI: 10.1109/TBME.2009.2030171.
- [8] M. Alwan et al. “A Smart and Passive Floor-Vibration Based Fall Detector for Elderly”. In: *2006 2nd International Conference on Information & Communication Technologies*. Vol. 1. 2006, pp. 1003–1007. DOI: 10.1109/ICTTA.2006.1684511.
- [9] Wendy Ju and Larry Leifer. “The Design of Implicit Interactions: Making Interactive Systems Less Obnoxious”. In: *Design Issues* 24 (July 2008), pp. 72–84. DOI: 10.1162/desi.2008.24.3.72.
- [10] Bellotti et al. “Making sense of sensing systems”. In: ACM Press, 2002. DOI: 10.1145/503376.503450.
- [11] Joe Marshall. *Joe’s Sensor Processing Interactive Teaching Stuff - Methods of combining sensors*. URL: https://joemarshall.github.io/websensors/lessons/6.Combining_Sensors/combining_3.html.
- [12] Joe Marshall. *grovepi-emulator - Emulator for GrovePI system*. URL: <https://github.com/joemarshall/grovepi-emulator>.
- [13] Henry Rimminen et al. “Detection of falls among the elderly by a floor sensor using the electric near field”. In: *IEEE Transactions on Information Technology in Biomedicine* 14.6 (2010), pp. 1475–1476. DOI: 10.1109/TITB.2010.2051956.
- [14] Jesse R. Codling et al. “FloHR: Ubiquitous Heart Rate Measurement using Indirect Floor Vibration Sensing”. In: *Proceedings of the 11th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation*. BuildSys ’24. New York, NY, USA: Association for Computing Machinery, 2024, pp. 44–54. ISBN: 9798400707063. DOI: 10.1145/3671127.3698170. URL: <https://doi.org/10.1145/3671127.3698170>.
- [15] Yohanna MejiaCruz et al. “Probabilistic Estimation of Cadence and Walking Speed From Floor Vibrations”. In: *IEEE Journal of Translational Engineering in Health and Medicine* 12 (2024), pp. 508–519. DOI: 10.1109/JTEHM.2024.3415412.
- [16] Marwa E. Ehaimir et al. “Human gait identity recognition system based on gait pal and pal entropy (GPPE) and distances features fusion”. In: *2017 Sudan Conference on Computer Science and Information Technology (SCCSIT)*. 2017, pp. 1–5. DOI: 10.1109/SCCSIT.2017.8293061.

A Instructions for Setup

A.1 Replaying data

Within the ‘replaying-data.py’ file, replace the ‘filepath’ variable to point towards your csv datafile, and edit the indexing to match the headers of the test file.

A.2 Live data

Connect an accelerometer sensor to i2c pin 3 and a sound loudness sensor to analogue pin 1 - or modify the ‘sensor-pins’ variable. Then run file ‘fall-detection-live.py’ using the GrovePi emulator or within the terminal.

B Sensor Probabilities

Acc. Sensor probabilities		
true positive rate	$P(A E)$	1.00
true negative rate	$P(A' E')$	0.93
false positive rate	$P(A E')$	0.00
false negative rate	$P(A' E)$	0.07

$P(E|A) \ 0.8$

Sound Sensor probabilities		
true positive rate	$P(S E)$	0.63
true negative rate	$P(S' E')$	0.66
false positive rate	$P(S E')$	0.38
false negative rate	$P(S' E)$	0.34

$P(E|S) \ 0.50$

(a)

(b)

Probs of sensor combinations						
both sensors true	$P(A \wedge S)$	0.18	$P(A \wedge S E)$	1	$P(A \wedge S E')$	0
	$P(A \wedge S')$	0.22	$P(A \wedge S' E)$	1	$P(A \wedge S' E')$	0
	$P(A' \wedge S)$	0.18	$P(A' \wedge S E)$	0.25	$P(A' \wedge S E')$	0.75
both sensors false	$P(A' \wedge S')$	0.42	$P(A' \wedge S' E)$	0	$P(A' \wedge S' E')$	1

Using Bayes Theorem + Prior.Prob

$P(E A \wedge S)$	0.8
$P(E A \wedge S')$	0.8
$P(E A' \wedge S)$	0.2
$P(E A' \wedge S')$	0

Using Bayes Theorem + Prior.Prob

$P(E A \wedge S)$	0.8
$P(E A \wedge S')$	0.8
$P(E A' \wedge S)$	0.2
$P(E A' \wedge S')$	0

(c)

Figure 17: Probabilities calculated from each sensors accuracy results, and combined using Bayes theorem. This gives the probability that a detected event corresponds to a true event, which has been used to decide the urgency of the alert raised in the case of each sensor combination [11].

C Results on Tile Surfaces



Figure 18: Testing results for the system on when placed on a tiled surface, including the raw data collected and the observed ground truth values. The accelerometer threshold can be adjusted to catch the missed event, while lowering the sound threshold will cause an increase in false-positives.

D Results on Laminate Surfaces



Figure 19: Testing results for the system on when placed on a laminate surface. The accelerometer has 100% accuracy, while the loudness sensor has missed all events due to the filter incorrectly interpreting the noisy environment.